FLORIAN BANON

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[LinkedIn]() • [GitHub]() • [Projects]()

Curious and driven **Computer Science and Aerospace Engineering** student completing my final internship as part of a dual Master's degree at Concordia University (Canada) and IPSA (France). Passionate in **machine learning, deep learning and satellite imagery processing**, I am now seeking a full-time opportunity to apply and further develop these skills.

SKILLS

Programming | Python • MATLAB • C++ • SQL • R

Skills | Machine learning • Deep Learning • AI • PyTorch • OpenCV • SAR Imagery • Git • Data Engineering

Methods | Engineering • Research • Communication • Autonomous • Proactive • Innovative • Thinking out of the box • Problem-solving • Project Management • Team building • Kaizen • 5S

Software | VSCode • Word • Excel • PowerPoint • MS Project • MS Teams • SIRIL • Amesim • MongoDB • SQLite STAR-CCM+ • ANSYS • CATIA V5 • SolidWorks • STK • Patran/Nastran

Languages | French – native • English – fluent

WORK EXPERIENCE

Research Engineering Intern in Machine/Deep Learning for SAR Imagery Processing **Jan 2026 – Present**
National University of Singapore, Singapore

- Developed a reproducible deep-learning pipeline in Python and PyTorch for data preparation, model training, evaluation, and experiment management.
- Designed and compared real-valued and complex-valued neural network approaches for learning from ionosphere-affected radar signal representations.
- Scaled the workflow to HPC environments to support larger experiments, with the long-term goal of improving the modelling, characterization, and mitigation of ionospheric effects in low-frequency SAR systems.

Dynamic Satellite Database Development Intern **Jun 2024 – Aug 2024**
ONERA, Palaiseau, France

- Utilized Python and MongoDB to build and optimize the database, implementing data pipelines for satellite tracking and analysis.
- Verified data integrity by developing scripts to detect and remove duplicates.
- Streamlining satellite cataloguing processes and improving consistency across datasets.

EDUCATION

Master of Engineering – Aerospace and Computer Science Engineering **2024 - 2026**
Concordia University, Montreal, QC

Master of Engineering – Aerospace and Computer Science Engineering **2022 - 2026**
Institut Polytechnique des Sciences Avancées, Paris, France

Exchange Semester **Oct 2023 – Mar 2024**
Universiti Sains Malaysia, Penang, Malaysia

PROJECTS

Dynamical Simulation of the Asteroid population
IPSA, Paris, France

- Modelled the motion of asteroid 2007 VW266 in retrograde resonance with Jupiter with Python.
- Analysed asteroid perturbations and approximated orbital elements.
- Simulated long-term asteroid trajectories to predict stability and future interactions with Jupiter (trajectory plot).

Neptune Flyby Mission
IPSA, Paris, France

- Calculated the trajectory and cost of a Neptune flyby mission, optimizing delta-V by integrating Jupiter flybys to minimize fuel consumption.
- Used NASA's trajectory browser to plan interplanetary transfers and simulated orbital manoeuvres with focus on hyperbolic escape and gravity assists.
- Delivered a full mission analysis covering orbit insertion, hyperbolic escape, and planetary encounter phases using STK.

VOLUNTEER WORK

Ropes Course Instructor **Jul 2018**
Karaez Adrenaline, Carhaix, France

INTERESTS

Reading	Fantasy, Philosophy, Manga, Comics, History
Sports	Bouldering (3 years), Tennis (5 years), Chess (1300 elo Glicko)
Passions	Hiking, Guitar, Piano